

Cinematic Insights: An Analysis of IMDB dataset using PCA and Clustering

Data Exploration & Summary

The IMDB movies dataset contains 50 records and 10 features for top-rated movies. It includes 5 categorical variables—Title, Genre, Description, Director, and Actors—and 5 continuous variables—Runtime, Votes, Revenue, and Year. We utilised all continuous variables except Year for this project. Among the variables, Revenue had 3 missing values. Consequently, we assessed its skewness (coefficient = 1.96). Given its positive skewness, we imputed missing values using the median. Moreover, we observed substantial variations in scale across numerical variables, leading us to standardise them to ensure consistency for Principal Component Analysis (PCA) analysis. Standardisation is crucial as it prevents the scale of variables from influencing PCA outcomes.

Section 1. Principal Component Analysis

PCA was conducted on scaled data to generate principal components (PCs) that account for different proportions of the variance within the dataset. This technique reduces the dimensionality of the data while preserving its essential features. It allows for transformation of high-dimensional data into a lower-dimensional space, facilitating easy interpretation.

Scree Plot Analysis

The scree plot in Figure 1 indicates the individual and cumulative explained variance by each PC. The first PC explains a substantial amount of variance, followed by a sharp drop, suggesting that the first PC captures the most significant patterns in the data. There is a levelling off after the second PC, with subsequent PC contributing less to the explanation of the variance.

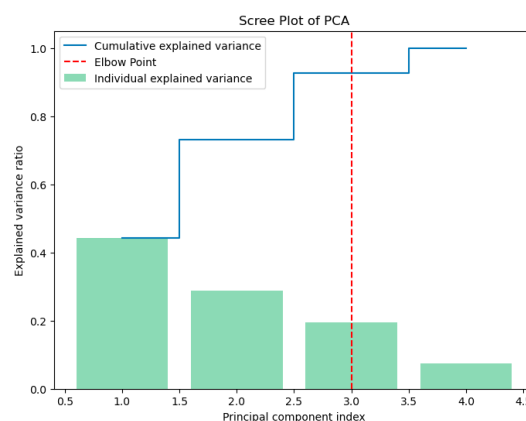


Figure 1. Optimal number of principal components – Scree Plot

Biplot Analysis

The biplot in Figure 2 combines a scatterplot of the scores of the principal components and the vectors of the original variables. It allows for the visualisation of the relationship between the variables and their contribution to each component. PC1 and PC2, the first two PC, show a distribution of the data points and the direction and magnitude of the original variables. Variables like 'Votes' and 'Revenue in Millions' have longer vectors & they are pointing in similar direction, indicating they are better represented by the first two PC and have a stronger positive correlation respectively.

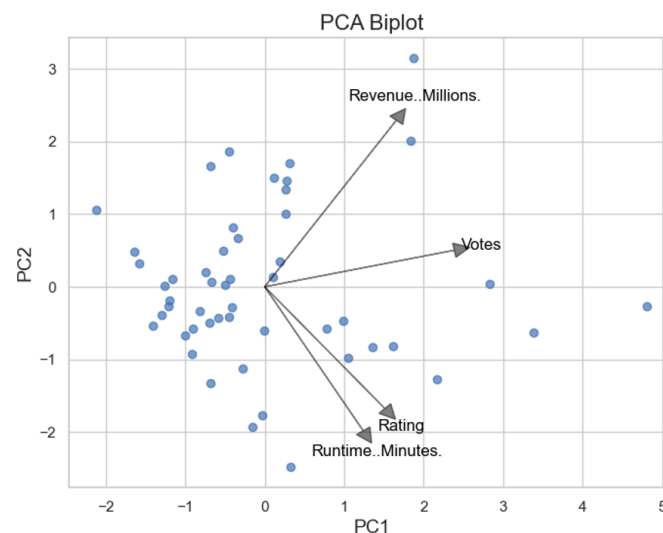


Figure 2. Feature variance projections – Bi Plot

Proportion of Variance Explained (PVE)

The PVE plot in Figure 3 shows a clear decline, which is typical in PCA, where the first few components explain most of the variance. The PVE for the first principal component is the highest, significantly dropping for the second and further for subsequent components.

Cumulative Proportion of Variance Explained:

The cumulative variance explained in an ascending plot in Figure 4, indicating how much of the total variance is captured by the addition of each principal component. With just the first two components, over 70% of the variance is explained, reaching approximately 90% by the third component, demonstrating that most information can be captured in three dimensions.

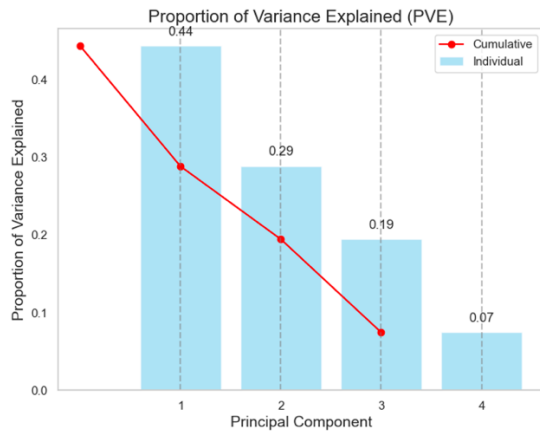


Figure 3. Key principal components explaining variance

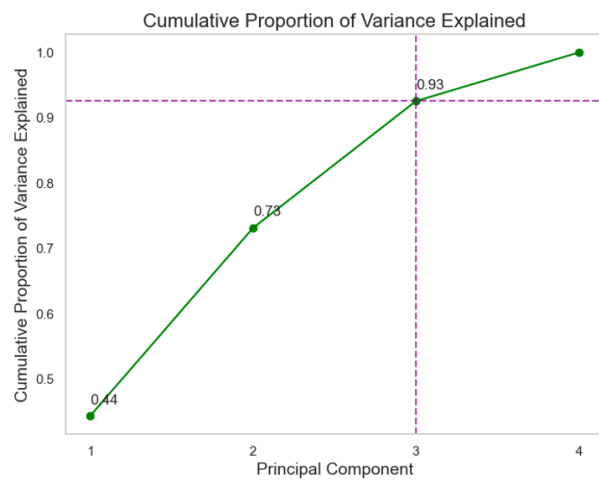


Figure 4. Cumulative proportion of variance explained

Singular Values

Singular values are standard deviations of PC, indicating variance captured by each PC. The first singular value is the largest, meaning that the first PC captures most variance in the dataset. This is consistent with the scree plot and PVE plot, where it shows a significant drop in explained variance after first PC. Table 1 represents the singular values of each PC.



Table 1. Singular values for component significance

PC	PC 1	PC 2	PC 3	PC 4
Singular Values	9.41	7.59	6.23	3.85

Key findings and results:

Analysis of the Scree plot, PVE plot, and Singular values suggests utilising the first three principal components to reduce dataset dimensionality, capturing over 90% of variance. This underscores the efficiency of concise data representation without significant information loss. Given the substantial variance explained by the initial 3 components, reducing dataset dimensionality is feasible. Therefore, we decided to move forward with 3 components for clustering analysis in next section.

Section 2. Clustering Methods

Following dimensionality reduction using PCA, we employed K-Means Clustering and Hierarchical Clustering methods to categorise movies according to their inherent similarities in numerical variables.

K Means Clustering

K-Means was chosen for its simplicity and efficiency in partitioning the dataset into K distinct, non-overlapping subgroups based on a pre-defined number of clusters. This approach was used to segment movies based on their PCA transformed features. To determine the optimal number of clusters, silhouette scores were computed for cluster sizes ranging from 2 to 10. The silhouette score assesses the similarity of an object to its own cluster compared to other clusters, with higher scores indicating better-defined clusters. The optimal number of clusters was identified as two, as it yielded the highest silhouette score, providing a quantifiable rationale for the choice. Figure 5 in Appendix A shows the distinct classes for K-Means Clustering.

Hierarchical Clustering (HC)

We employed HC as an alternative method to visualise cluster arrangement through a dendrogram. Using the 'ward' method for linkage, we aimed to minimise variance within clusters. Determining the distance threshold to cut the tree and define the clusters involved identifying the point where merging clusters no longer provides significant gains in reducing within-cluster variance. We selected this by identifying the largest gap in the dendrogram, marked by the longest vertical line not intersected by any horizontal lines. This led us to

choose a distance threshold of 7, intersecting with one vertical line, resulting in 3 clusters. Figure 6 and 7 in Appendix A displays the dendrogram and distinct classes for hierarchical clustering respectively.

Key findings and results:

The optimal number of clusters determined for the K-Means algorithm was 2, with a silhouette score of approximately 0.40. This suggests that the data points within each cluster are, on average, closer to each other than to points in other clusters. The dendrogram from hierarchical clustering suggests different linkages between movies based on their features, leading to three main clusters with a silhouette score of approximately 0.34. when cutting at a distance of 7.

Cluster Profile for K-Means & Hierarchical Clustering (Table 2 and 3 in Appendix B)

Table 2 for K-means represents Cluster 0 with longer movies, high ratings, high votes, and revenues, while Cluster 1 includes less popular short films with lower ratings, fewer votes and significantly lower revenues. Table 3 for Hierarchical Clustering represents Cluster 1 containing high-performing, widely recognised movies, whereas Cluster 2 films features with highest average revenue despite having lower average ratings and votes. This could be due to the inclusion of several high-grossing animation and family movies, as indicated by common genres. Cluster 3 is the largest and most diverse, encompassing movies with lower average revenue and votes, indicating varied performance compared to Clusters 1 and 2.



Section 3. Conclusion & Limitations

Our analysis provides a comprehensive view of grouping movies based on features such as ratings, revenue etc. PCA highlights revenue and votes as key contributors to variance among top-rated movies. K-Means clustering yielded two distinct groups separating highly popular, high-grossing movies from those with average performance, supported by silhouette scores which offer an objective evaluation of cluster quality. This supports our decision for adoption of K-Means as the preferred clustering method. However, hierarchical clustering uncovers subtle distinctions, including high-revenue yet potentially lower-rated films, overlooked by K-Means. Future analyses could explore advanced clustering techniques that handles mixed data types incorporating genre and director information, alongside temporal analysis for trends over time.

Appendix A: K-Means and Hierarchical Clustering Plots

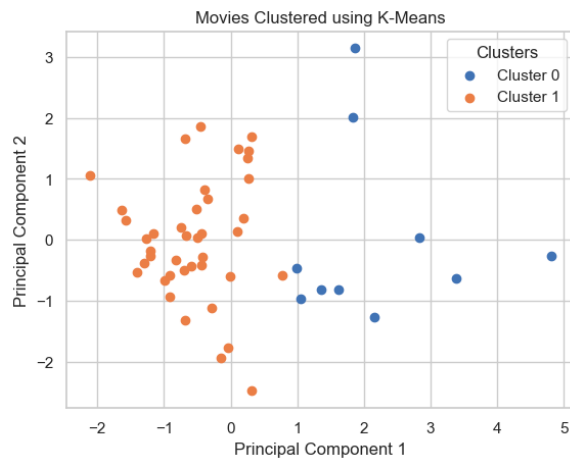


Figure 5 K-Means Clustering

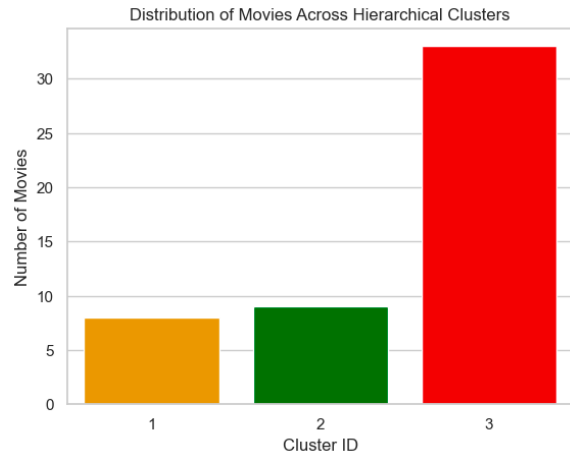


Figure 6 Hierarchical Clusters

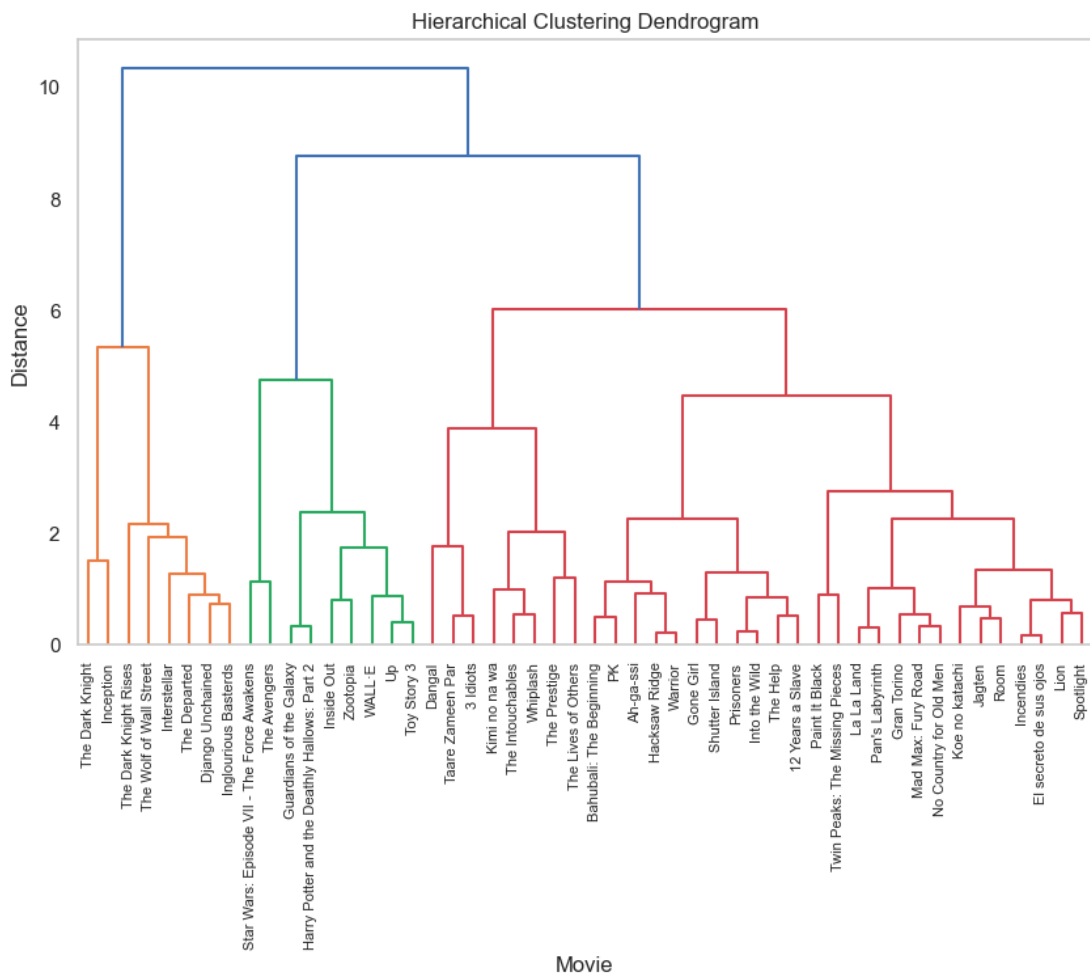


Figure 7. Hierarchical Clustering - Dendrogram

Appendix B: Clustering Profiles for K-Means and Hierarchical Clustering

Table 2. Cluster Profiles for K-Means

Cluster ID	Average Runtime (min)	Average Rating	Average Votes	Average Revenue (Millions)	Number of Movies	Most Common Genres	Most Common Directors
0	156.1	8.4500	1,115,385.7	355.448	10	Action, Crime, Drama; Action, Adventure, Sci-Fi	Christopher Nolan (4)
1	127.525	8.2575	361,614.1	102.137	40	Drama (5); Animation, Adventure, Comedy (4)	Damien Chazelle (2)

Table 3. Cluster Profiles for Hierarchical Clustering

Cluster ID	Average Runtime (min)	Average Rating	Average Votes	Average Revenue (Millions)	Number of Movies	Most Common Genres	Most Common Directors
1	160.25	8.5375	1,180,833	249.32125	Size of Cluster 1	Action, Crime, Drama; Action, Adventure, Sci-Fi	Christopher Nolan (4)
2	114.444	8.1889	650,464	433.72	Size of Cluster 2	Animation, Adventure, Comedy (4)	Pete Docter (2)
3	131.818	8.2667	312,654.1	52.78515	Size of Cluster 3	Drama (5); Drama, Mystery, Romance (2)	Damien Chazelle (2)